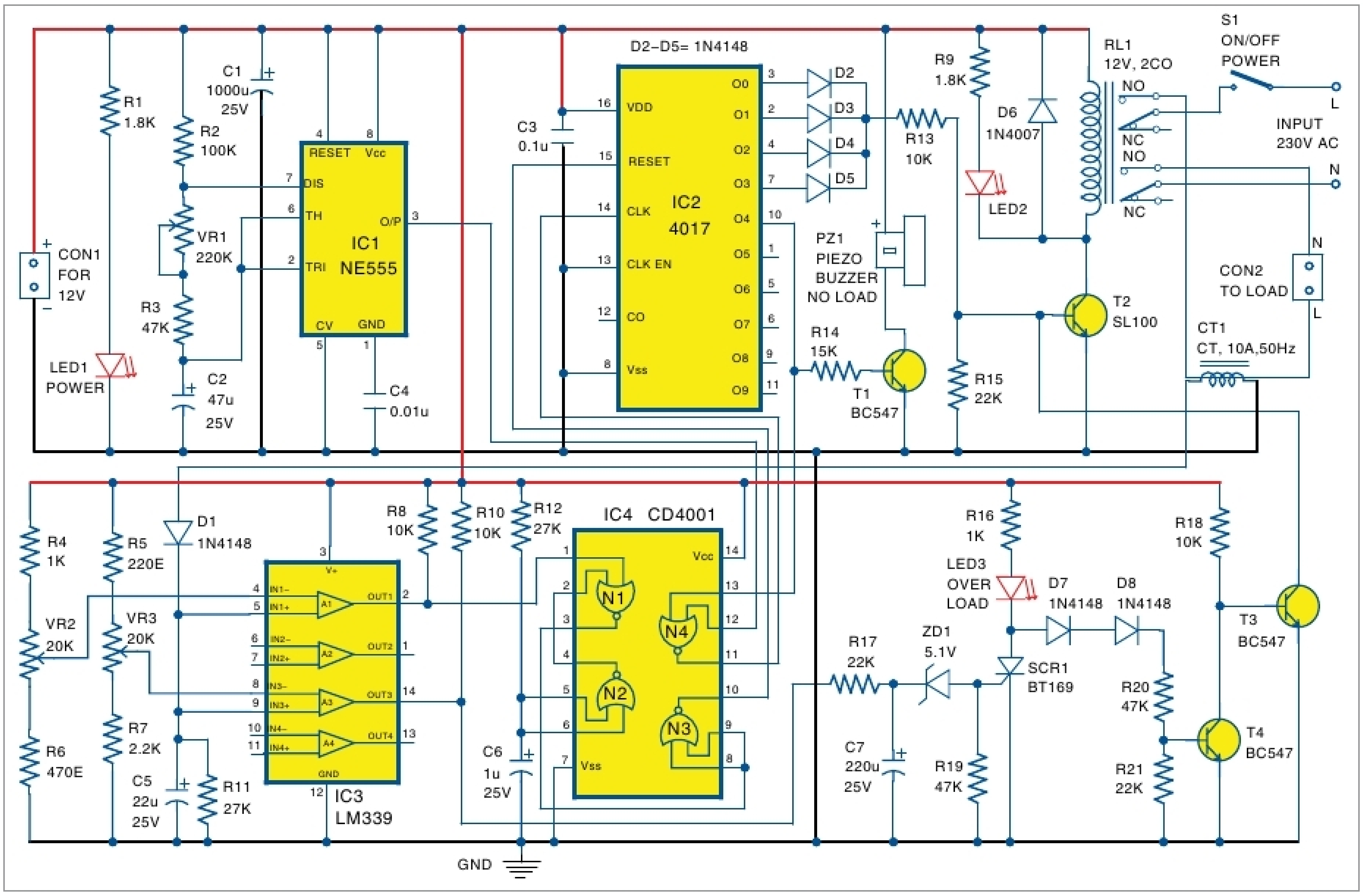

Fig. 2: Circuit diagram of no-load and overload protector for AC motor

for no-load and overload conditions, accordingly.

Output at pin 2 of IC3 is high, and output at pin 14 is low (after initial reset time delay). N1 output at pin 3 of IC4 is low and inverted output at pin 10 of N3 is high. This high condition keeps IC2 in reset state. Q0 is high and counter IC2 does not progress. High Q0 is used to switch on T2 and energise RL1. This connects AC mains power to the load through relay contacts. In this condition, T3 is cut off and maintains base bias of T2 as high.

No-load or low-load current condition. In this condition, DC sample voltage from the rectifier at pin 5 is less than the reference voltage at pin 4 from VR2. Output of A1 at pin 2 of IC3 becomes low and output of N1 at pin 3 of IC4 becomes high. Inverted output at pin 10 of N3 becomes low. Reset pin 15 of IC2 becomes low. This allows the counter to progress through Q1, Q2, Q3 and Q4. Here, Q0 to Q3 are ORed and used to drive T2, so that T2 conducts till counter output reaches Q4.

This mechanism is provided to establish sufficient delay before cutting off the supply to the load. This delay can be adjusted by changing the frequency of astable multivibrator IC1 by adjusting the value of potmeter VR1. Once Q4 becomes high and drives T1, the piezo buzzer (PZ1) is switched on. The high Q4 is fed to pin 13 of NOR gate N4, which blocks the clock pulses of IC2 at pin 14. Functioning of counter IC2, logical circuits and comparators freezes till next switching off and on of AC mains supply using switch S1.

If 12V SMPS is used, its input supply should be taken through switch S1.

Overload or short-circuit condition. In this condition, DC sample voltage from rectifier at pin 9 goes more than the voltage across VR3. Output of A3 at pin 14 changes from low to high. This high level is applied across reverse-biased Zener diode ZD1 through an integrator formed by resistors R10 and R17, and capacitor C7. This integrator network and Zener diode provide sufficient time delay to fire SCR1 through its gate. Time delay is required to avoid unwanted tripping off the load by initial in-rush current to inductive load or occurrence of any brief overload condition.

In case SCR1 is fired due to prolonged (more than delay time imposed by integrator and Zener diode) overload condition, anode potential of SCR goes less than one volt. This small anode voltage is not sufficient to forward bias diodes D7 and D8, which, in turn, cut off T4. Cut off condition of T4 makes T3 switch on, and its collector potential to come down to ground potential. This, in turn, pulls down base bias of T2, leading it to cut off. This leads to de-energising RL1 and withdrawal of mains supply to load.

Once SCR1 is fired and starts conducting between cathode and anode, status continues till the power switches off and on again. LED3 (initially dim) glows bright, indicating the registered status of overload condition till next restart. Withdrawal of power supply to load by RL1 establishes a no-load condition. The no-load sensing circuit starts to act, and audio alarm sounds